

School of Computer Science Engineering and Technology

Lab No. - 20

Course-B. Tech.	Type- Core
Course Code- CSET301	Course Name- Artificial Intelligence and Machine Learning
Year- 2025	Semester- Odd
Date-	Batch- 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
Q1						

Lab – Implementation of K-Means Algorithm

Total Marks: 1

Objective:

To implement the K-Means Clustering algorithm using Scikit-learn on the MNIST dataset, visualize clusters in reduced dimensions using PCA, and evaluate clustering performance using appropriate metrics.

Problem Statement

The objective of this lab is to implement K-Means Clustering on the MNIST handwritten digit dataset using Scikit-learn. The MNIST dataset consists of 70,000 grayscale images of handwritten digits (0–9), each represented as a 28×28 image (784 features). Since K-Means is an unsupervised learning algorithm, it attempts to group similar data points into clusters without using the digit labels during training.

In this task, you will:

- Perform data preprocessing by flattening the images and applying standardization using StandardScaler.
- Apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset to 2 components for better visualization.
- Use the K-Means algorithm to cluster the PCA-reduced data into 10 groups, ideally representing the 10-digit classes.
- Visualize the clustered data in 2D space using a scatter plot, with each point coloured by its predicted cluster label.

- Evaluate the clustering performance by comparing predicted cluster labels with the true digit labels using a confusion matrix and accuracy score, optionally aligning clusters with true labels using the Hungarian Algorithm.
- Interpret the clustering results to understand the challenges and effectiveness of using unsupervised learning techniques like K-Means on high-dimensional image data.

Download the dataset from:

<https://www.kaggle.com/c/digit-recognizer/data>
 (Alternative source:) `tensorflow.keras.datasets.mnist`

About Dataset:

The MNIST dataset consists of 70,000 grayscale images of handwritten digits (0–9). Each image is 28x28 pixels (784 features). This dataset is widely used for training image processing systems and serves as a standard benchmark in machine learning.

Attribute Information:

- Input features: 784-pixel values per image
- Target: Digit label (0–9)
- Type: Unlabelled (for clustering), labels used for evaluation only

Questions:

1. Data Pre-processing:
 - Load the MNIST dataset.
 - Flatten the 28x28 images into 784-length vectors.
 - Normalize the data using StandardScaler.
2. Dimensionality Reduction:
 - Reduce the dataset to 2 dimensions using PCA for visualization.
 - Use PCA-transformed data as input for K-Means clustering.
3. K-Means Clustering:
 - Apply K-Means clustering with `n_clusters=10`.
 - Fit the model on PCA-reduced data.
 - Assign cluster labels to each sample.
4. Visualization:
 - Plot the PCA-reduced 2D data, color-coded by predicted cluster labels.
 - Optionally, visualize cluster centers.
5. Evaluation:
 - Compare predicted cluster labels with true digit labels using:
 - Confusion matrix
 - Accuracy using `accuracy_score`